

Cryptography - 24_10

Proof

Let $GAME_{\Pi, \mathcal{A}}^{ufcma}(\lambda) \equiv G(\lambda)$ and $H(\lambda)$ is the same but replace $F_k(\cdot)$ with $R \leftarrow \mathcal{R}(\lambda, n \rightarrow n)$.

Standard reduction to PRF security gives $G(\lambda) \approx_c H(\lambda)$.

It remains to prove: $\forall PPT \mathcal{A}$

$$\Pr[\underbrace{\mathcal{A} \text{ wins in } H(\lambda)}_{H(\lambda)=1}] \leq \text{negl}(n)$$

Indeed, this probability is $\leq 2^{-n}$ for even unbounded $\mathcal{A}$.

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The question now is: How about longer messages? How about VIL?

Let's start with some warm-up examples: Let $m = m_1, \dots, m_d$

- $\tau_i = \text{Tag}(K, m_i)$, $\forall i \in [d]$ (d blocks), with $\tau = (\tau_1, \dots, \tau_d)$
Doesn't work! Attacker can mix and match: For instance $\text{Tag} ? m_1 || m_2$ then $m_3 || m_4$, forge $\tau_1 || \tau_4$ on $m_1 || m_4$.
- $\tau = \text{Tag}(k, \bigoplus_{i=1}^d m_i)$
Even this one is not secure: Here is an attack. For instance, query $m || m$ for any $m \in \{0, 1\}^n$. This gives $\tau = \text{Tag}(K, 0^n)$.
Now output: $m' || m' = m^*$ and $\tau^* = \tau$, with $m' \neq m$.

Idea: Let $\mathcal{H} = \{h_s : \{0, 1\}^{n \cdot d} \rightarrow \{0, 1\}^n\}_{s \in \{0, 1\}^\lambda}$.

Then, let $\text{Tag}(k, m) = F_k(h_s)$.

What property of $\mathcal{H}$ is enough? Clearly, it should be hard to find m, m' such that $m \neq m'$ and $h_s(m) = h_s(m')$ (a collision). This property is known as collision resistance.

✓ Definition (AU = Almost Universal)

$\mathcal{H}$ is $\mathcal{E}$ - almost universal if $\forall m, m'$ such that $m \neq m'$

$$\Pr_s[h_s(m) = h_s(m')] \leq \mathcal{E}$$

$\mathcal{E} = 2^{-n}$ is called Perfectly Universal

$\mathcal{E} = \text{negl}(|s|)$ is AU.

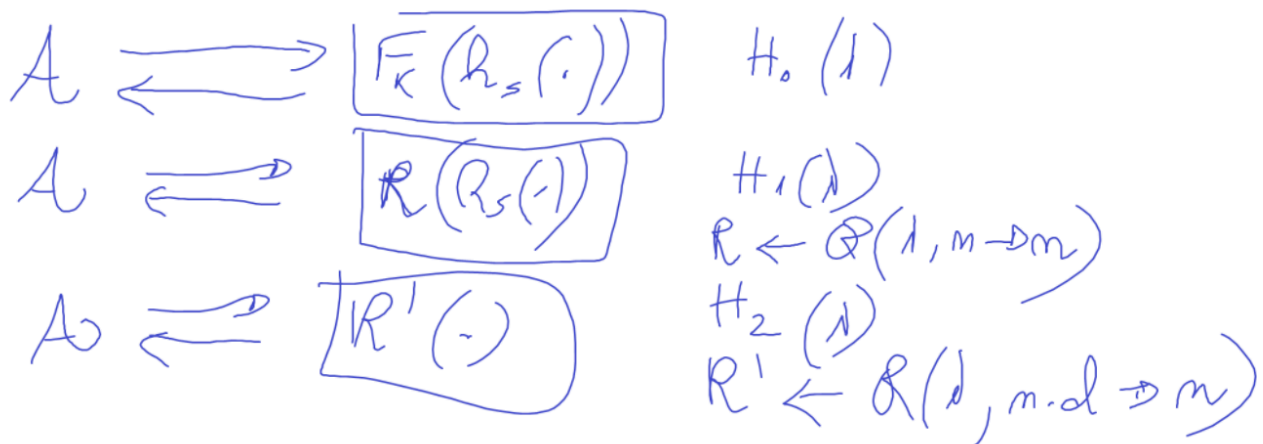
✓ Theorem

If $\mathcal{F}$ is a PRF, and $\mathcal{H}$ is AU, then $\mathcal{F}(\mathcal{H})$ is a PRF.

Where $\mathcal{F}(\mathcal{H}) = \{F_k(h_s(\cdot))\}$

📌 Proof

We consider a few experiments:



So, three hybrid worlds.

Standard reduction: $H_0(\lambda) \approx_c H_1(\lambda)$.

We have that A_{01} makes a query $x \in \{0, 1\}^{n.d}$, then A_{prf} samples $s \in U_\lambda$ and pass $h_s(x)$ to C_{prf} . It answers by sending back y , passed then to A_{01} and then it distinguish b' .

It remains to show: $H_1(\lambda) \approx_c H_2(\lambda)$. We actually show they are $\approx_s$ so long as # queries = $q(\lambda) = \text{poly}(\lambda)$.

Let BAD: the event that becomes true if $\exists i, j$ among $x_1, \dots, x_q$ such that $i \neq j$ and $h_s(x_i) = h_s(x_j)$. Conditioning on $\overline{\text{BAD}}$, $H_1(\lambda) \equiv H_2(\lambda)$. So,

$$SD(H_1; H_2) \leq \Pr[\text{BAD}].$$

Subtlety! We first sample s and then answer the queries. Whereas AU requires to first fix the queries and then sample s (that's the problem).

So, we redefine BAD: Upon input x from $\mathcal{A}$ reply with U_n . At the end, sample s and check if $\exists i, j$ such that $h_s(x_i) = h_s(x_j)$. However, nothing changed. Until BAD happens, the two are identical. After, we don't care.

So we have that

$$\begin{aligned}
Pr[BAD] &= Pr_s[\exists x_i, x_j \text{ s.t. } h_s(x_i) = h_s(x_j)] \leq \sum_{i,j; i \neq j} Pr[h_s(x_i) = h_s(x_j)] \leq \\
&\leq \binom{q}{2} \cdot \mathcal{E} = \text{negl}(\lambda)
\end{aligned}$$

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It's easy to construct AU families. Two examples:

$$\begin{aligned}
1. \quad &F = GF(2^n); m = m_1, \dots, m_d, \text{ with } m_i \in GF(2^n), \text{ and} \\
&s = (s_1, \dots, s_d) \in (GF(2^n))^d, \text{ and } h_s(m) = \sum_{i=1}^d s_i \cdot m_i = \langle \vec{s}, \vec{m} \rangle.
\end{aligned}$$

Take $m = (m_1, \dots, m_d)$ and $m' = (m'_1, \dots, m'_d)$, with $s_i = m'_i - m_i$.

$$\begin{aligned}
\text{Then } h_s(m) = h_s(m') &\iff \sum_{i=1}^d s_i \cdot m_i = \sum_{i=1}^d s_i \cdot m'_i \\
&\iff s_1 \cdot \delta_1 = - \sum_{i=2}^d s_i \cdot \delta_i \text{ (i take out the first difference)} \\
&\iff s_1 = \frac{- \sum_{i=2}^d s_i \delta_i}{\delta_1}
\end{aligned}$$

Now the key is really long, as long as the message.

$$\implies \mathcal{E} = 2^{-n}$$

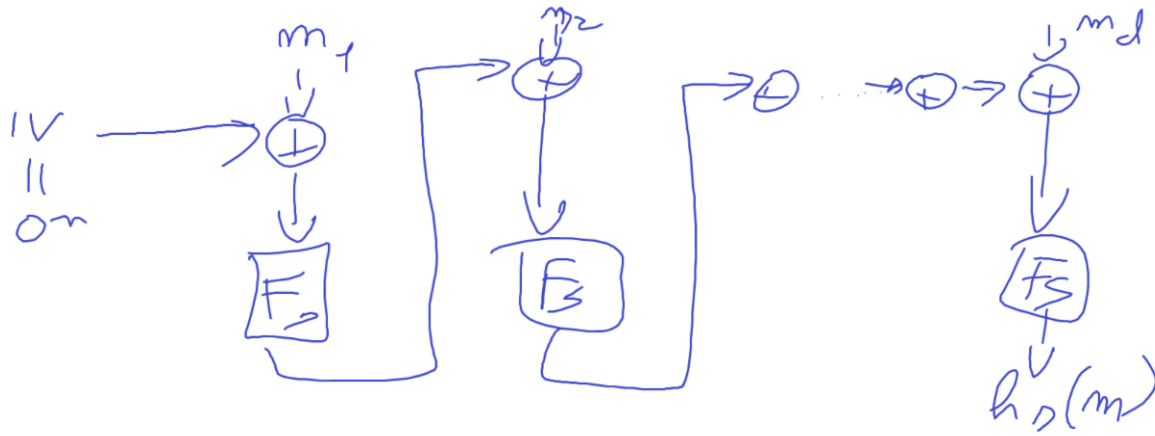
$$2. \quad F = GF(2^n); s \in F$$

$$h_s(m) = h_s(m_1, \dots, m_d) = \sum_{i=1}^d m_i \cdot s^{i-1} = q_m(s)$$

Now $h_s(m) = h_s(m')$

$$\begin{aligned}
&\iff q_m(s) = q_{m'}(s) \\
&\iff q_{m-m'}(s) = 0 \\
&\iff \sum_{i=1}^d (m_i - m'_i) \cdot s^{i-1} = 0 \\
&\implies \mathcal{E} = \frac{d-1}{|F|} \approx \frac{d}{2^n} = \text{negl}(\lambda)
\end{aligned}$$

CBC-MAC: It's just a different choice of h_s , using "another" PRF $F_s(\cdot)$.



The difference with standard CBC is that IV is 0^n . Also, we don't output at each step

✓ Theorem

The above $\mathcal{H}_{CBC}$ is computationally AU.